Mental Health in Tech

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***Abstract*— Psychological, emotional and social well-being make up a healthy state of mind. Mental health related issues have been a great matter of concern in almost every field of work and life. It is said that at any given time, 1 in 5 adults are living with mental illness. But, OSMI's Mental Health in Tech Survey suggests that the number is higher in tech. Predicting such issues in the early stages will greatly help the patient to avoid it or receive the right treatment for it. Here, we aim to apply data analytics on the data related to mental health in tech to derive some insights from the data, obtain answers to some valuable questions and make some predictions on the patient's mental health. Selection of appropriate data analytic techniques also plays an important role here. Thus, the objective of this study is, early detection of any signs of possible mental health issues of people in tech. The process includes, (i)performing data analytics on the data to understand and analyze the data with respect to different features, (ii)predicting whether or not the subject will be diagnosed with a mental issue by classifying the subject and (iii)clustering the subject into one of the 3 risk levels(high, medium, low), where the risk levels refer to the risk of the subject showing any signs of mental health issues. The results thus obtained can be further used to prevent the occurrence of mental health issues and assist in providing quality mental health services to the patients.**

Keywords— ***mental health, data analytics, classification, prediction, mental issues in tech***

I. Introduction

WHO defines mental health as mental well-being in which an individual realizes his or her own abilities, can cope with the normal stresses of life, can work productively and is able to make a contribution to his or her community. We know that technology has made our lives easier and allowed us to work more efficiently. As much as it has influenced us positively, it also has negatively affected the lives of people working in the tech industry. According to OSMI's data, 51% of tech professionals have been diagnosed with a mental health condition. Thus, it becomes important to study information related to mental health in tech and identify the main factors influencing the subject's mental health. It is also necessary to analyze the data, identify hidden patterns and encapsulate it into useful information.

The emerging field of 'predictive analytics in mental health' has recently generated tremendous interest with the bold promise to revolutionize clinical practice in psychiatry paralleling similar developments in personalized and precision medicine [7]. Prediction in data mining involves attributes or variables in the data set to find an unknown or future state values of other attributes [9].

But we realize that just predicting if the subject will be affected by a mental health issue doesn't provide adequate information about the subject's mental health. It is also important to study the patterns of the subjects who are diagnosed with mental issues and that of those who are not diagnosed with any. Their similarities and differences must be analyzed to detect any signs of possible mental issues of the subjects.

Thus, in this paper, the predictions as to whether or not the subject will be diagnosed with a mental health issue is compounded with another indicator which describes the risk of the subject getting affected by a mental health issue. The risk factor is described in 3 levels, high, medium and low risk. Each of the subjects fall into one of the 3 clusters, which allows us to understand the subjects, their environments and sometimes the factors which influence the subject's mental health.

II. Literature Survey

A. *Statistical procedures for analyzing mental health services data- Jon D. Elhai, Patrick S. Calhoun , Julian D. Ford [1]*

There is a large body of work that has examined variables associated with previous use of mental health services using conceptual frameworks [2]. Numerous works have concluded that this data is significantly associated with demographic characteristics, attitudes towards treatments, mental health diagnoses and access variables [3]][4][5]. However, there are some methodological issues to this problem due to design specific problems in querying about service use and in measuring utilization [6]. This is

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worsened by numerous data analysis issues that warrant consideration.

The paper by Elhai, Calhoun and Ford gave us valuable insights into some commonly used statistical procedures for analyzing mental health services data. The paper presented a review of some common data analysis problems that are inherent while analyzing data from mental health services and surveys. There is always a potential danger of using analytic methods whose assumptions are violated by the nature of data distribution. Mental health services use studies with very complexly distributed data and therefore, the assumptions made by any model must be made to match the data distribution. The paper lists some of its complexities and describes the cost data. It also details some useful data transformations and regression models. Some of the solutions listed in the paper with a vision to maximize utility and accuracy include,

1) Poisson and negative binomial
2) Zero-inflated regression model
3) Zero-truncated regression model

The aim of the paper was to examine assumptions made by previous methods that reduce data utility. Relying on logistic or linear methods or data transformations are found to be likely to over or understate the actual variance and may reverse the direction. These results can often mislead researchers in detrimental ways. When studies suggest stronger relationships by specific predictors with mental health illnesses, utilizers may falsely assume that the type of illness or treatments should be geared to address that specific predictor. For example, if the study finds higher data utilization with men, it may be interpreted as indicating that women are under-utilizing services. The paper also highlighted how spurious associations can obscure meaningful relationships. When the data analyses under-estimates overall associations, the importance of predictors may be under-valued. This may lead to false assumptions. Therefore, the selection of appropriate data analytic techniques is very important to our statistical matter.

B. *Predictive Analytics in mental health: applications, guidelines and perspectives- T Hahn, AA Nierenberg, S Whitfield-Gabrieli[7]*

One of the main objectives of this paper is to provide an overview of the important questions and problems faced in the field of predictive analysis in mental health. It aims to propose some general guidelines for predictive analytics projects in psychiatry.

It states that predictions of individual risks are very important with respect to both short-term and long-term predictions. It could largely increase the efficiency of preventive treatments as all the efforts could be focused only on the individuals that are at risk.

Thus, valid models in predictive analysis could be very effective in not only reducing the sufferings of the patient but also increasing the efficient allocation of resources for research.

As we all know that any predictive analytics project includes the following four major steps, i.e. defining the objective of the model, acquiring data for the model, building the model and using the model in clinical practice. Each of the above steps is crucial and influences the model. Thus, this paper provides some guiding questions which help in the analytics and also address the factors that are needed to ensure the validity and applicability of a model. For example, whether or not the prediction of an unknown variable is needed; will there be any relationship between predictor variables and the predicted variables; scope of the model etc.

The paper also addresses some of the challenges that could occur during analytics like curse of dimensionality, modelling massively multivariate and multimodal data etc. Some solutions include dimensionality reduction techniques such as PCA, techniques integrating dimensionality reduction and predictive model estimation (for example, regularization, Bayesian model selection and cross-validation) and feature engineering to extract all the useful predictors from the input data. It also mentions model integration techniques such as stacking and voting that can be used to have the most efficient combination of data. Thus, the emphasis on asking the right questions about the data and the model, handling the challenges that occur during the process of analytics and lastly translating the research findings into practice effectively which needs both domain knowledge and data integration technology is learnt here.

C. *A Novel approach for Psychiatric Patient Detection and Prediction using Data Mining Techniques-Dr. E. Chandra Blessie, Bindu George.[8]*

The main aim of this paper is to study the effective use of classifiers to develop an intelligent mental disorder diagnosis and prediction system. Here, comparative evaluation of Naïve Bayes and K-Nearest Neighbor classifiers is done in the psychiatric patient dataset. The evaluation of the for the same is done. After studying the evaluations provided, the aim is to choose a suitable classifier which effectively selects the influential features and accurately provides the mental disorder diagnosis and prediction.

In this paper, it is stated that the care for mental health is majorly based on self-assessment as the mental issues are the consequences of the patients' behavior in their lives. Thus, data mining algorithms which have a good classification accuracy will provide greater amounts of information to detect the patients with mental issues. It also explains that with the help of a predictive model, we can recognize the patient which requires more care and concern as compared to other patients.

Here, two classifiers, Naive Bayes algorithm and K-Nearest Neighbor algorithm are used for classification and comparison. Accuracy, sensitivity and specificity are used for comparison of results.

The paper then concludes on the note that algorithms that are developed to extract information and discover certain patterns in the data are useful for decision making. It also hints that the extracted data patterns can provide useful information to prevent mental illness and assist in delivery of efficient mental health services.

III. Dataset

OSMI's Mental Health in Tech Survey 2016 is used in this study. The 2016 survey aims to measure attitudes towards mental health in the tech workplace and examine the frequency of mental health disorders among tech workers. It consists of over 1400 responses collected for 63 questions related to mental health of the respondents, their view towards mental health in the workplace, their awareness of mental health, their demographics etc.

IV. Exploratory Data Analysis

Most of the data available to us fit into the following categories:

1) Ordinal data
2) Interval data
3) Nominal data

Before conducting data analysis, we renamed all columns to easily referenceable names that were uniform. Data preprocessing was done starting with data cleaning which included handling missing values, values with the same meaning represented in different forms and outliers. Plotting heat maps gave us a clear visual sense on the data and sparsity. The columns which had more than 50% missing (or Nan) values were dropped

With a view to provide inclusivity, the dataset had a non-categorical column for gender. However, with an aim to provide uniformity, and to generate insights relating to the subject's sex, we decided to condense all 70 unique values in that column to just 3, i.e. female, male and other. The age column is critical in all health-related surveys, so it was imperative that we ensure that all outliers were removed and replaced by the mean of all the ages.

There were a few fields that implied very similar categories, elicited response biases and fields whose entries were directly dependent on prior fields. For example, the missing values in the columns which were related to the subject's previous employer and previous workplace were due to the absence of a previous employer for the subject. Due to this dependency, the missing values in all such columns were replaced with another value 'Not applicable'. In an effort to provide flexibility in opinion and explanation, there were two columns in the survey that allowed the subjects to elaborate and provide comments. While this information is certainly valuable for studies into the field of mental health, for the sake of visualization analytics, we decided to drop the two columns. The values of countries with long names were replaced with acronyms like 'USA' for 'United States of America' for easier representation. It was seen that the subjects who were self-employed had missing values in the corresponding rows of the column 'Total_employees'. Hence, the missing values in that column were imputed with the value '1-5' as most of the self-employed subjects had a total of 1-5 employees. The missing values in the column 'Tech_organization' were replaced by the mode of the same column. It was also seen that the values observed in the remaining columns were dependent upon whether or not the subject was self-employed. Thus, each of the remaining columns was grouped by the column 'Self_employed' (which contained 2 unique values 0 and 1). Modes were calculated for each column with respect to each group and all the missing values in that column were imputed with the corresponding modes.

Finally, the rows which still contained missing (or Nan) values were dropped.

Cleaning was followed by encoding the categorical columns to prepare the dataset for the model. It was observed the values in most of the categorical columns were also ordinal. Thus, ordinal encoding was done on the categorical columns so as to retain the ordinal nature of the features. For example, 'Awareness_of_options_under_coverage' column consisted of 3 unique values i.e. 'Yes', 'No' and 'I am not sure'. These 3 unique values were encoded as 2,1 and 0 respectively. The column 'work_pos' described one or more position of the subject in their workplace like: Supervisor/Team Lead|Support|Back-end Developer. This column was replaced by the column 'No_of_pos' indicating the number of positions held by the subject in their workplace.

The columns 'us_state_live', 'us_state_work', 'country_live' and 'country_work' had the names of various states of the USA. All the states were divided into 4 divisions called: 'north_east', 'mid-west', 'west' and 'south' and encoded as 1,2,3 and 4 respectively.

Basic correlation analysis and chi square tests were performed on columns to gain useful insights on the subjects as well as possible ideas for visualization.

Correlation between the columns 'premp_provide_resources_learn_seekhelp' and 'premp_took_menhealth_seriously_as_phyhealth' was found. This helps to understand if employers providing resources to their employees to learn about mental issues and seek help for the same has an effect on the attitude of the employer regarding mental issues. Chi square test done on the same pair of columns indicates that the 2 columns are indeed related to each other. The correlation coefficient was a positive value 0.257 which hints that providing employees with resources to learn about mental health and seek help

for mental issues does indicate that the employer takes mental health as seriously as physical health.
Similarly, chi square test done on the columns 'Mental_health_coverage' and 'Comfort_discussing_with_supervisor' indicates that the 2 columns are related to each other. The correlation coefficient found was a small positive value 0.076 which hints that there can be a slight chance that an employee provided with mental health benefits by their employer finds it comfortable to discuss mental health related issues with their direct supervisors.

Lastly, chi square test done on the columns 'aware_benefits_prev_empl' and willing_discuss_issue_dir_supervisors' also indicates that the 2 columns are related to each other. The correlation coefficient found was a small positive value 0.03 which hints that there can be a slight chance that a subject with a mental issue can be willing to discuss the same with their co-workers if they are aware of the mental health benefits provided at their workplace.

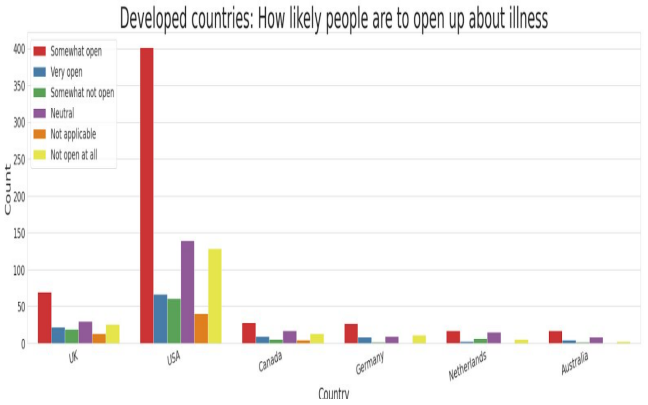
Our survey had a total of 53 countries. Among these, the majority of respondents were found to be from the US, UK, Canada, Germany, Netherlands and Australia.
Generally, for a sample to be representative enough for the population, the size needs to be sufficiently high (at least >30 for our data). While we could have decided to avoid using all countries that had fewer than 30 respondents, this would have significantly restricted the scope of our project to big economies with high living standards. In an effort to include a variety of nations to compare and contrast their conditions, we decided to group the nations based on a few classes to reach an adequate number of countries per class:

1) Based on region (Europe, Asia Australia and Africa, and America)
2) Based on economic development

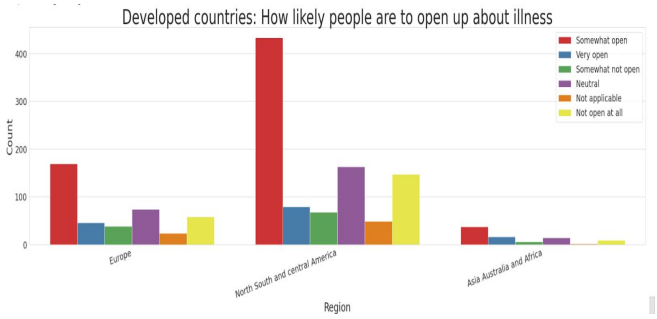
The above measures were necessary for us to determine which tools would be most suited to visualize our data.

V. Data Visualization

1) *How likely people are to opening up about mental health illnesses -- country and region wise analysis*

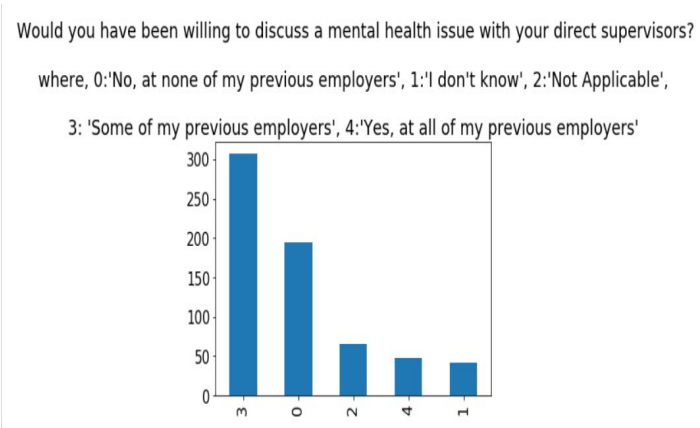


The bar charts clearly indicate that most of the country's employees are not likely to open up about their mental health issues. The USA has a much higher count of employees who are open which is a good sign but still there are a fair amount of people who are not fine with discussing their mental health. Many countries like Canada, UK, Germany, Netherlands, Australia have an incredibly low amount of people opening up, but this could also be due to very less amount of data collected from these countries.



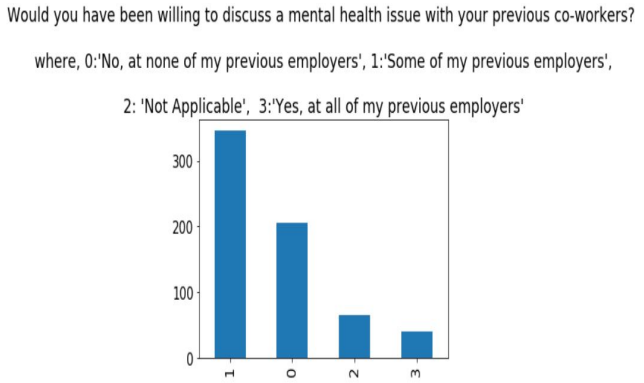
Analyzing the same region wise, we find that people in North, South and Central America are relatively more open about mental issues as compared to Europe which is followed by regions of Asia, Australia and Africa which show the least likeliness of people opening up about their mental issues.

2) *How willing are the employees to discuss their mental health issue with their direct supervisors?*



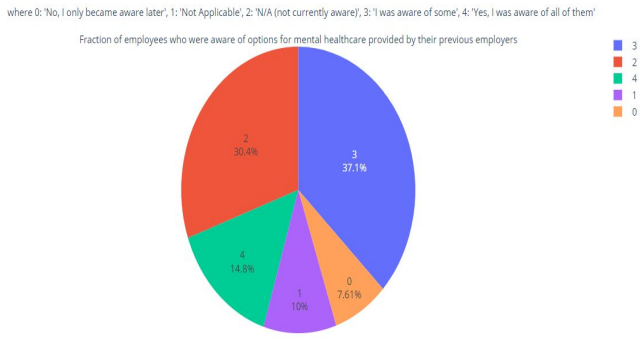
Most of the employees find it easier to talk to their previous direct supervisors about their mental issues, while some still do not discuss with some of their previous supervisors.

3) *Willingness of the employees to discuss their mental health issues with their coworkers at previous employers' workplace.*



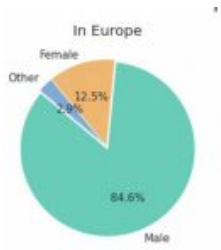
Most of the employees find discussing their mental health with their previous co-workers comfortable, while a considerable number of them do not find it comfortable.

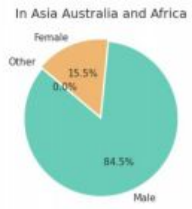
4) *Fraction of employees who were aware of options for mental health care provided by their previous employers.*

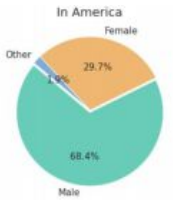


From the visualization it can be seen that 37.1% of the employees were aware of only some of the options provided by their previous employers, while a large 30.4% of the employees were not aware of any of the options and only about 14.8% of them were aware about all of the options. This clearly shows that the companies must take the initiative to make all the employees aware of all their options under mental health.

5) *Gender proportions in the workplace*



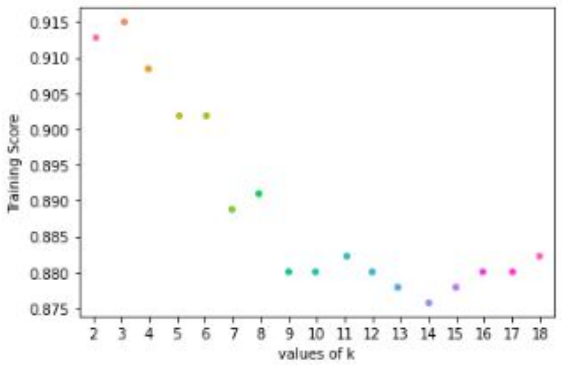


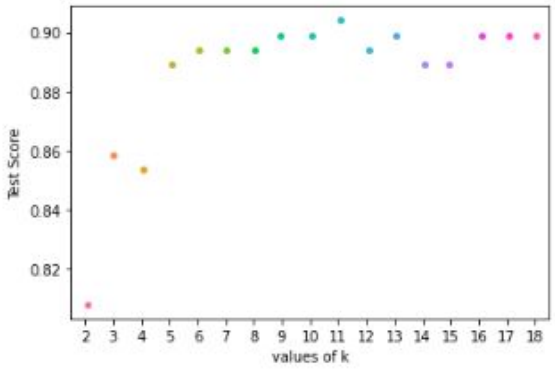


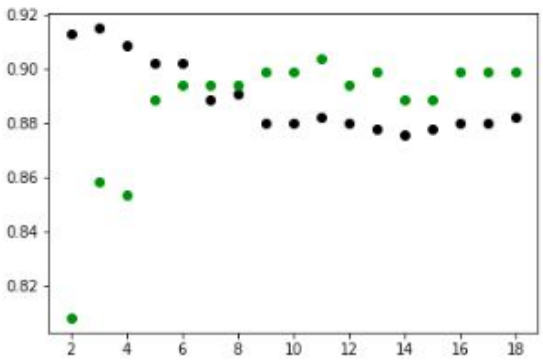
The pie charts clearly show that most of the employees in the tech are men. America comparatively has a higher proportion of female to male.

VI. Classification

K-nearest neighbors (KNN) classification technique is used to classify the subjects into 2 classes namely: 'diagnosed for mental health issue by a professional' and 'not diagnosed for mental health issue by a professional'. Thus the target variable for classification is the column 'diagnosed_prof'. The dataset is split into train and test sets, the train set consisting of 70% of the entire data and the train set consisting of 30%. Both test and train sets are standardized using StandardScaler which transforms the dataset such that its mean (μ) = 0 and standard deviation (σ) = 1. The target variable used was whether or not a person has been diagnosed of mental health related illness professionally. By plotting train and test scores with different values of K, we were able to find the optimum number of neighbours for the classifier.







We obtained a test accuracy of 89.89%. The output is whether a person would or would be diagnosed as having a mental health illness. While a classification of this nature gives users unambiguous clarity, the intention is not to give the user a binary prediction.

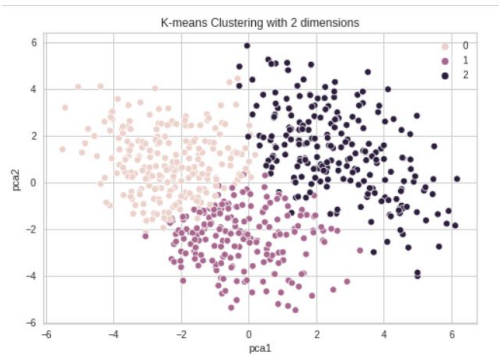
VI. Clustering

K means clustering technique is used to cluster similar instances of the data. This allows us to understand the subjects, their environments and sometimes the factors which influence the subject's mental health. Again, the preprocessed dataset is standardized using StandardScaler transforming the dataset such that its mean (μ) = 0 and standard deviation (σ) = 1.

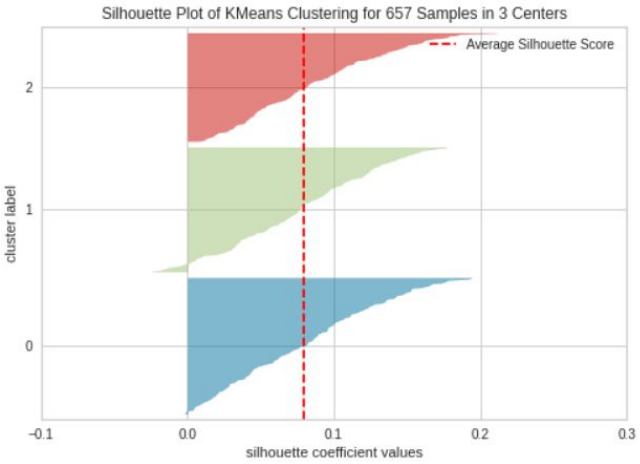
K means clustering is done iteratively for k = 1,2,3…6 clusters using sklearn library. Within Cluster Sum of Squares scores (WCSS) are calculated for each of the 6 iterations and the same is visualized using a graph. It can be seen that the elbow point is obtained at k=3. Similarly, Calinski Harabasz scores are calculated for each of the 5 iterations 2,3,4,5,6 and are plotted. This graph also indicates the elbow point at k=3. Thus, the optimum number of clusters for this dataset it obtained as 3. K means is then run on the dataset with k=3.

The clusters thus obtained are plotted using a scatter plot by reducing the number of features to 2 using PCA (Principal Component Analysis).

The scatter plot shows that the model has rightly clustered similar instances of the data.



Various evaluation metrics are calculated to validate the results. The Silhouette score, which ranges from (-1, 1) where 1 indicates dense clusters and -1 indicates sparse clusters, was found to be equal to an acceptable value of 0.079. A graph indicating the same is plotted.

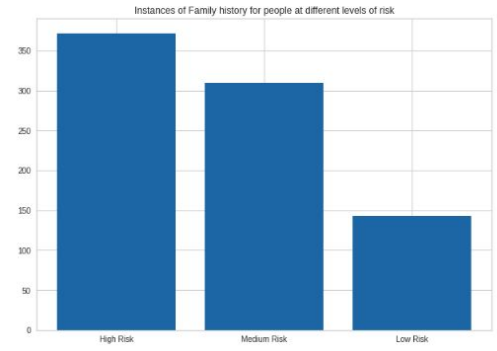


The Sum of squared errors (SSE) was calculated as 26202.171 and the Calinski Harabasz score was calculated as : 58.366.

Each of the 3 clusters are then identified as the clusters indicating the risk of a subject getting affected by a mental health issue. This risk is defined in 3 levels namely: high, medium and low. The number of subjects who have been diagnosed with a mental issue is calculated for each of the 3 clusters. The cluster with the highest number of subjects diagnosed with a mental issue is labelled as the high risk cluster, the cluster with the next highest number of subjects diagnosed with a mental issue is labelled as the medium risk cluster and lastly the cluster with the least number of subjects diagnosed with a mental issue is labelled as the low risk cluster.

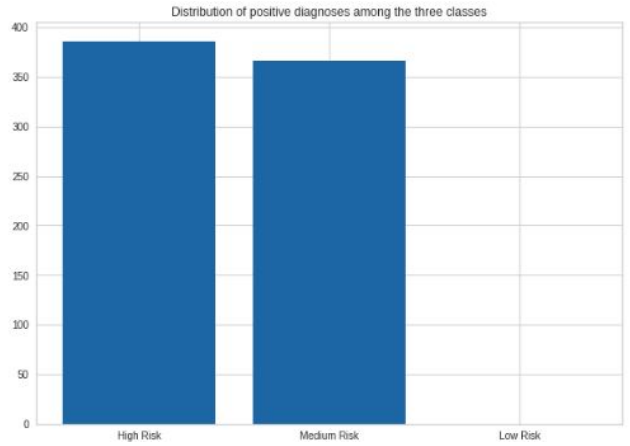
VII. Inferences

Using the three classes of risks, valuable insights could be made based on the other features of people who fall in the three different clusters

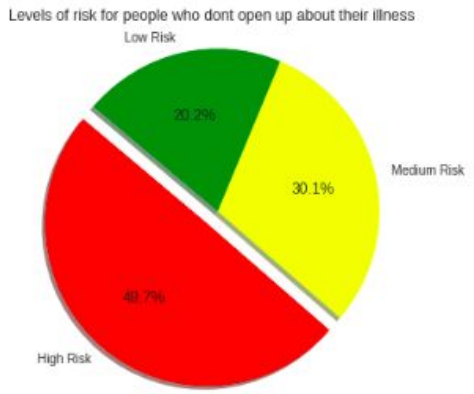


Upon analyzing family history in the different clusters, we found that people who fall in high risks were also the ones who showed

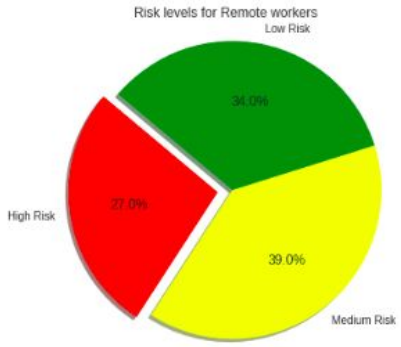
higher cases of family history with illness which supports the belief that family illness is likely to affect future generations.

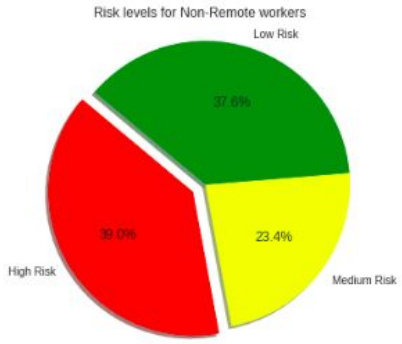


We then analyzed people who do not open up to friends and family about their illness and found that close to 50% of such cases were more likely to be at a high risk of mental health illness. This goes to show that having a support system can go a long way in helping people confront their problems and that conversation should be encouraged.



Another analysis conducted was within the context of the pandemic. We expected to find remote working to be lonely; putting people at a higher risk. Surprisingly, we found that the bigger proportion of high risk- 39%, was found in non-remote workers. We then set out to analyze why this may be the case

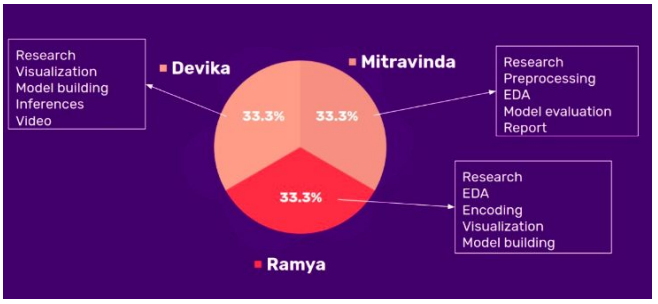




In our most profound insight so far, we analyzed the effect of negative behavior towards mental health in the workplace. We found that a staggering 65% employees who experienced negative reactions from their workplace were at a higher risk of mental health illnesses.

This goes to show that we still have a long way to go in our treatment and understanding of mental health. Providing support for mental health in the workplace is important to debase the stigma attached with its discussion. Using machine intelligent assessment tools like ours in offices and schools is a positive step in that direction.

VIII. Contributions



In terms of contribution to our project, each of us put our maximum effort and our contribution to our project is reflective of this as well- Mitravinda worked on data preprocessing, EDA, model evaluation and our meticulously crafted report. Ramya worked on EDA, visualization, model building and research. Devika worked on visualizing, model building, risk calculation and video.

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